Year 10 Higher

REVISION PACK

Graphs 2

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 2 · Distance-time graphs (Lesson 8.2)

Q	Working	Answer
time stopped and speed home		
1.1	$35 - 20 = 15$ $8 \text{ km in } 20 \text{ min}$ $8 \div \frac{1}{3} = 24$	**(a) 15 minutes** **(b) 24 km/h**
1.2	$60 - 40 = 20$ $10 \text{ km in } 30 \text{ min}$ $10 \div \frac{1}{2} = 20$	**(a) 20 minutes** **(b) 20 km/h**
1.3	$45 - 15 = 30$ $6 \text{ km in } 15 \text{ min}$ $6 \div \frac{1}{4} = 24$	**(a) 30 minutes** **(b) 24 km/h**
1.4	$50 - 30 = 20$ $9 \text{ km in } 30 \text{ min}$ $9 \div \frac{1}{2} = 18$	**(a) 20 minutes** **(b) 18 km/h**

Graphs 2 · Speed-time graphs (Lesson 8.3)

Q	Working	Answer
acceleration and total distance		
2.1	$(a)\ 10 \div 5 = 2$ $(b)\ \frac{1}{2} \times 5 \times 10 + 10 \times 10 + \frac{1}{2} \times 5 \times 10$ $= 25 + 100 + 25 = 150$	**(a) 2 m/s²** **(b) 150 m**
2.2	$(a)\ 12 \div 4 = 3$ $(b)\ \frac{1}{2} \times 4 \times 12 + 6 \times 12 + \frac{1}{2} \times 6 \times 12$ $= 24 + 72 + 36 = 132$	**(a) 3 m/s²** **(b) 132 m**

2.3	$(a)\ 18 \div 6 = 3$ $(b)\ \frac{1}{2} \times 6 \times 18 + 10 \times 18 + \frac{1}{2} \times 4 \times 18$ $= 54 + 180 + 36 = 270$	**(a) 3 m/s²** **(b) 270 m**
2.4	$(a)\ 20 \div 10 = 2$ $(b)\ \frac{1}{2} \times 10 \times 20 + 10 \times 20 + \frac{1}{2} \times 5 \times 20$ $= 100 + 200 + 50 = 350$	**(a) 2 m/s²** **(b) 350 m**

Graphs 2 · Perpendicular lines (Lesson 8.4)

Q	Working	Answer
equation of a perpendicular line		
3.1	$m = -\frac{1}{2}$ $1 = -\frac{1}{2} \times 6 + c$ $c = 4$ $y = -\frac{1}{2}x + 4$	$y = -\frac{1}{2}x + 4$
3.2	$m = -\frac{1}{4}$ $5 = -\frac{1}{4} \times 8 + c$ $c = 7$ $y = -\frac{1}{4}x + 7$	$y = -\frac{1}{4}x + 7$
3.3	$m = \frac{1}{3}$ $4 = \frac{1}{3} \times 3 + c$ $c = 3$ $y = \frac{1}{3}x + 3$	$y = \frac{1}{3}x + 3$
3.4	$m = -\frac{1}{5}$ $3 = -\frac{1}{5} \times 10 + c$ $c = 5$ $y = -\frac{1}{5}x + 5$	$y = -\frac{1}{5}x + 5$

3.5	$m = \frac{1}{2}$ $7 = \frac{1}{2} \times 4 + c$ $c = 5$ $y = \frac{1}{2}x + 5$	$y = \frac{1}{2}x + 5$
3.6	$m = -\frac{1}{3}$ $0 = -\frac{1}{3} \times 6 + c$ $c = 2$ $y = -\frac{1}{3}x + 2$	$y = -\frac{1}{3}x + 2$

Graphs 2 · Mid-points and length of a line (Lessons 8.5, 8.6)

Q	Working	Answer
mid-point and length		
4.1	**(a)** $\left(\frac{-1+5}{2}, \frac{2+10}{2}\right)$ **(b)** across $= 6$, up/down $= 8$ $AB = \sqrt{6^2 + 8^2} = \sqrt{100}$ $= 10$	**(a) (2, 6)** **(b) 10**
4.2	**(a)** $\left(\frac{-3+9}{2}, \frac{4+(-1)}{2}\right)$ **(b)** across $= 12$, up/down $= 5$ $AB = \sqrt{12^2 + 5^2} = \sqrt{169}$ $= 13$	**(a) (3, 1.5)** **(b) 13**
4.3	**(a)** $\left(\frac{2+6}{2}, \frac{-3+4}{2}\right)$ **(b)** across $= 4$, up/down $= 7$ $AB = \sqrt{4^2 + 7^2} = \sqrt{65}$ $= 8.1\ldots$	**(a) (4, 0.5)** **(b) 8.1**

4.4	**(a)** $\left(\frac{-1+2}{2}, \frac{-1+3}{2}\right)$ **(b)** $\text{across} = 3,\ \text{up/down} = 4$ $AB = \sqrt{3^2 + 4^2} = \sqrt{25}$ $= 5$	**(a) (0.5, 1)** **(b) 5**
4.5	**(a)** $\left(\frac{0+5}{2}, \frac{-4+8}{2}\right)$ **(b)** $\text{across} = 5,\ \text{up/down} = 12$ $AB = \sqrt{5^2 + 12^2} = \sqrt{169}$ $= 13$	**(a) (2.5, 2)** **(b) 13**
4.6	**(a)** $\left(\frac{-6+2}{2}, \frac{1+7}{2}\right)$ **(b)** $\text{across} = 8,\ \text{up/down} = 6$ $AB = \sqrt{8^2 + 6^2} = \sqrt{100}$ $= 10$	**(a) (−2, 4)** **(b) 10**